MILITARY SPECIFICATION

PACKAGING OF DRIVE, BALLISTIC T23 AND T23E1

1. SCOPE

1.1 This specification covers the packaging, packing and marking for domestic shipment and storage, overseas shipment, and interplant shipment of Ballistic Drive T23 and T23E1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standard of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

NN-B-601	-	Boxes, Wood-Cleated-Plywood, For Domestic Shipment.
NN-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
QQ-S-781	-	Strapping, Flat; Steel.
UU-C-843	-	Cushioning Materials, Cellulosic.
UU-P-313	-	Paper, Lens Tissue.
LLL-F-291	-	Fiberboard; Corrugated, Single Face (Flexible).

Military

JAN-P-105	-	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106	-	Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-108	-	Packaging and Packing for Overseas Shipment - Boxes; Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116	-	Preservation, Methods Of.
JAN-P-117	-	Packaging and Packing for Overseas Shipment - Bags, Interior Packaging.
JAN-B-121	-	Barrier-Materials, Greaseproof.
JAN-P-127	-	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure Sensitive, Water-Resistant.
JAN-P-139	-	Packaging and Packing for Overseas Shipment - Plywood, Container Grade.

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Military (contd.)

MIL-A-140	-	Adhesive, Water-Resistant, Waterproof, Barrier Material.
MIL-B-131	-	Barrier Material, Water-Vaporproof, Flexible.
MIL-L-644	-	Lubricating-Oil, Preservative, Special.
MIL-L-3150	-	Oil, Lubrication, Preservative, Medium.
MIL-C-5092	-	Cement, Rubber (Synthetic and Reclaimed Rubber Base).
MIL-R-6855	-	Rubber; Synthetic, Sheet, Molded and Extruded, for Aircraft Application.
MIL-C-11796	-	Compound Corrosion Preventive, Petrolatum-Type, Hot Application.

Ordnance Department Tentative

AXS-1472	-	Adhesive, Label (Water-Resistant).

STANDARD

Military

MIL-STD-129	-	Marking of Shipments.

(Copies of specifications and standards required by the contractor in connection with specific procurement requirements should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials - The following shall apply wherever packaging materials are referred to in this specification by name only:

Reference	Full Identification
Cellulosic cushioning material	Any type and class (at least 1/4 inch thick) of Specification UU-C-843
Flexible single face corrugated fiberboard	Type I of Specification LLL-F-291
Greaseproof bag	Type II or III, grade B, class a of Specification JAN-P-117
Greaseproof barrier material	Type II or III, grade C, class 1 of Specification JAN-B-121
Neutral tissue paper	Type II of Specification UU-P-313
Preservative compound	Class 3 of Specification MIL-C-11796
Preservative oil	**Specification MIL-L-3150**

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Reference	Full Identification
Pressure sensitive tape	Any type and grade of Specification JAN-P-127
Water-vaporproof barrier material	Class 1 of Specification MIL-B-131

3.2 Disassembling and cleaning

3.2.1 Disassembling - The linkage shall be disassembled from the ballistic drive unit at the coupling. The linkage on Ballistic Drive T23E1 shall then be disassembled by removing the rod which joins the connector and the lower connector assembly together; also, the connector shall be disconnected from the arm assembly. The linkage on Ballistic Drive T23 shall be disassembled by removing the rod which joins the connector assembly and the lower connector assembly together.

3.2.2 Cleaning - The exterior surfaces of each ballistic drive shall be cleaned so as to be free from dirt, smudges, fingerprints and other foreign substances in accordance with cleaning methods C-1 and C-4 of Specification MIL-P-116. Method C-4 shall be used to clean all critical unprotected surfaces such as mounting surfaces.

3.3 Packaging - **(See figures 3, 4, and 5 for packaging of a ballistic drive.)**

3.3.1 Arm Assembly - Preservative conforming to Specification MIL-L-3150 shall be applied on the mounting hardware (screws and washers) for the mounting support and coupling. The hardware shall then be placed in a greaseproof bag which shall be secured to the arm assembly with pressure sensitive tape. Preservative (compound) shall be applied on the mounting surfaces of the mounting support and coupling. Preservative shall be applied on the screw used to secure the connector (on Ballistic Drive T23E1) to the connector assembly attached to the arm. The screw shall be turned into the connector assembly until it is as far as it will go. All coated surfaces shall be wrapped or covered with greaseproof barrier material. Where it is necessary, the barrier material shall be secured with pressure sensitive tape, then overwrapped with flexible single face fiberboard and secured with pressure sensitive tape.

3.3.2 Rod - Preservative compound shall be applied on the threaded ends of the rod. The coated surfaces shall be wrapped with greaseproof barrier material which shall be secured in place with pressure sensitive tape, then overwrapped with flexible single face fiberboard and secured with pressure sensitive tape.

3.3.3 Lower connector assembly - Preservative compound shall be applied on the stud of the lower connector assembly. The coated surfaces shall be wrapped with greaseproof barrier material which shall be secured in place with pressure sensitive tape, then overwrapped with flexible single face fiberboard and secured with pressure sensitive tape.

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3.3.4 Ballistic unit assembly

3.3.4.1 Preservative oil shall be applied on the mounting hardware and on unpainted surfaces, except the coupling which shall be coated with a preservative conforming to Specification MIL-L-644. The coated mounting hardware shall be placed in a greaseproof bag and secured to the ballistic unit with pressure sensitive tape. Other surfaces coated with a preservative shall be covered or wrapped with greaseproof barrier material, which shall be secured in place with pressure sensitive tape where necessary.

3.3.4.2 The level vial shall be covered with the metal cover provided. The metal cover shall be secured in place with pressure sensitive tape.

3.3.4.3 The glass surface of the ballistic unit shall be covered with at least two thicknesses of neutral tissue paper. At least two thicknesses of cellulosic cushioning material shall be placed over the tissue covered glass surface and secured in place with pressure sensitive tape.

3.3.4.4 The base assembly of the ballistic unit shall be wrapped with cellulosic cushioning material in such a manner that the screw heads are cushioned. The wrap shall be secured with pressure sensitive tape. The body of the ballistic unit shall then be overwrapped with flexible single face fiberboard and secured with pressure sensitive tape.

3.3.4.5 Each ballistic unit assembly shall be inside a floating bag and bolted to the base shown in figure 1.

3.3.4.5.1 The floating bag shall be fabricated from flexible water-vaporproof barrier material. Sufficient barrier material shall be provided at the opening of the bag to make two additional seals. The opening of the bag shall be heat sealed in such a manner that the bag can be opened and resealed at least twice. A section of flexible water-vaporproof barrier material, approximately 10" x 15", shall be centered on the inside of the floating bag and heat sealed to the bag, as shown in figure 4. Gaskets fabricated from barrier material conforming to class II, grade 60 of Specification MIL-R-6855 shall be cemented to the blocks (shown in figure 4) and to the barrier material with cement conforming to type 2 or 3 of Specification MIL-C-5092.

3.3.4.5.2 Four 3/8" x 2-1/4" carriage bolts shall be used to secure the ballistic unit assembly to the base as shown in figure 5. Flat steel washers and steel lock washers shall be added under the nuts. Unthinned paint shall be applied on the exposed threads above the nuts. Asphalt sealing compound or water-resistant adhesive, conforming to Specification MIL-A-140, shall be applied over the bolt heads and on a portion of the wood around the bolt heads.

3.3.4.5.3 The floating bag shall be completely sealed after exhausting entrapped air.

3.3.4.6 The packaged items shall be secured to the wood base by means of flat steel straps as shown in figure 6. The

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straps shall conform to type I, class A or B of Specification QQ-S-781. Where a connector for Ballistic Drive T23E1 is involved, the connector shall be secured to the wood base in the same manner.

3.4 Packing - **(See figure 6 for packing the Ballistic Drive.)**

3.4.1 For domestic shipment and storage and overseas shipment

3.4.1.1 The wood base containing the ballistic drive shall be placed in a shipping container conforming to either of the following specifications:

JAN-P-105, Style A or B. Plywood shall conform to type A or B, condition I of Specification JAN-P-139
JAN-P-106, Style 2 or 4. Minimum thickness of lumber for sides, top and bottom shall be 11/16 inch, when shipment is to be made by an ocean going vessel

3.4.1.2 Size and weight of packed shipping container - The following inside dimensions, cubage and gross weight are approximate for a nailed wood box:

Inside dimensions - 26" long x 17-1/2" wide x 14" deep
Cubage - 3.7 cubic feet
Gross weight - 91. pounds

3.4.1.3 The steel straps shall not be stapled on the shipping container for domestic shipment and storage.

3.4.2 Immobilizing - The wood base shall fit snugly in the shipping container and shall be immobilized by nailing a hold down cleat in each corner of the container as shown in Figure 6. Each hold down cleat shall be at least 12-1/4" long x 1-1/2" wide x 25/32" thick.

3.5 Identification and marking

3.5.1 In addition to other markings that may be required (see 6.2), the information to be placed on each container shall be in accordance with Standard MIL-STD-129. The information shall be stencilled or printed on the container. The serial number of the ballistic drive in the container shall be added in the identification data. (See 6.3 for the stock number to be added in the identification data.)

3.5.2 Official Ordnance Corps Inspection Seal (Escutcheon) - Either a paper seal or a stamped escutcheon shall be placed adjacent to the permanent markings on each shipping container. The Ordnance District's symbol, such as NYOD, BOD, etc., shall be added on each paper seal or adjacent to each stamped escutcheon. If paper seals are used, they shall be affixed and coated with water-resistant label adhesive conforming to Specification AXS-1472.

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3.6 Interplant shipment requirements - Where an interplant shipment is involved, the following cleaning, packaging, packing and marking requirements shall apply:

3.6.1 Disassembling and cleaning - Disassembling and cleaning shall be in accordance with 3.2.2.

3.6.2 Packaging - Packaging shall be as specified in 3.3, except as follows:

3.6.2.1 Application of preservatives may be omitted.

3.6.2.2 The floating bag, gaskets and application of adhesive over the bolt heads may be omitted. The finished mounting surfaces of the mounting supports shall be covered with greaseproof barrier material.

3.6.3 Packing

3.6.3.1 The shipping container shall conform to the following specifications:

NN-B-601 NN-B-621	JAN-P-108, Compliance symbol V1 or V2 and any style, except FTC

3.6.3.2 If a fiberboard box is used, the framework shown in Figure 2 shall be nailed to the wood base. The base and framework shall fit snugly in the fiberboard container. The fiberboard box shall be closed and sealed as specified in the box specification.

3.6.4 Identification and marking - Identification and marking shall be as specified in 3.5, except that the serial number may be omitted.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling - A preproduction sample pack shall be made in accordance with this specification. The sample pack shall be inspected and approved by a Government inspector prior to packaging and packing the item for shipment. Production packs shall be tested at random where the Government inspector considers it necessary to conduct tests to assure conformance with the requirements of this specification.

4.2 Inspection - All methods and materials shall be subject to inspection by a Government inspector, who shall be afforded proper facilties for determining compliance with this specification and the applicable specifications.

4.3 Test procedures - The preproduction sample pack and any production packs selected for tests shall, at the discretion of the Government inspector, be tested in accordance with the applicable tests specified in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

(Not applicable.)

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6. NOTES

6.1 Ordering data - Invitation for bids and contracts or orders should specify the title, number and date of this specification, the type of shipment involved, and the type of packaging and packing required.

6.2 The marking of overseas shipping containers by Army Installations should be in accordance with Department of the Army and Air Force Manual SR 746-30-5, "Marking of Oversea Supply".

6.3 The following indicates the stock number for each ballistic drive which is to be added in the identification data:

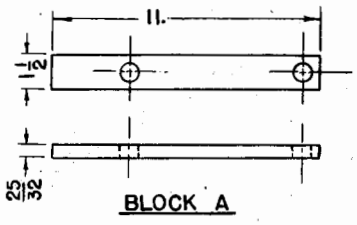
Drive, Ballistic T23	F001-7649132
Drive, Ballistic T23E1	F001-7649131

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

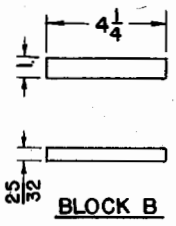
Custodian:

Army - Ordnance Corps

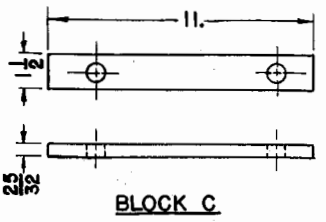
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BLOCK A

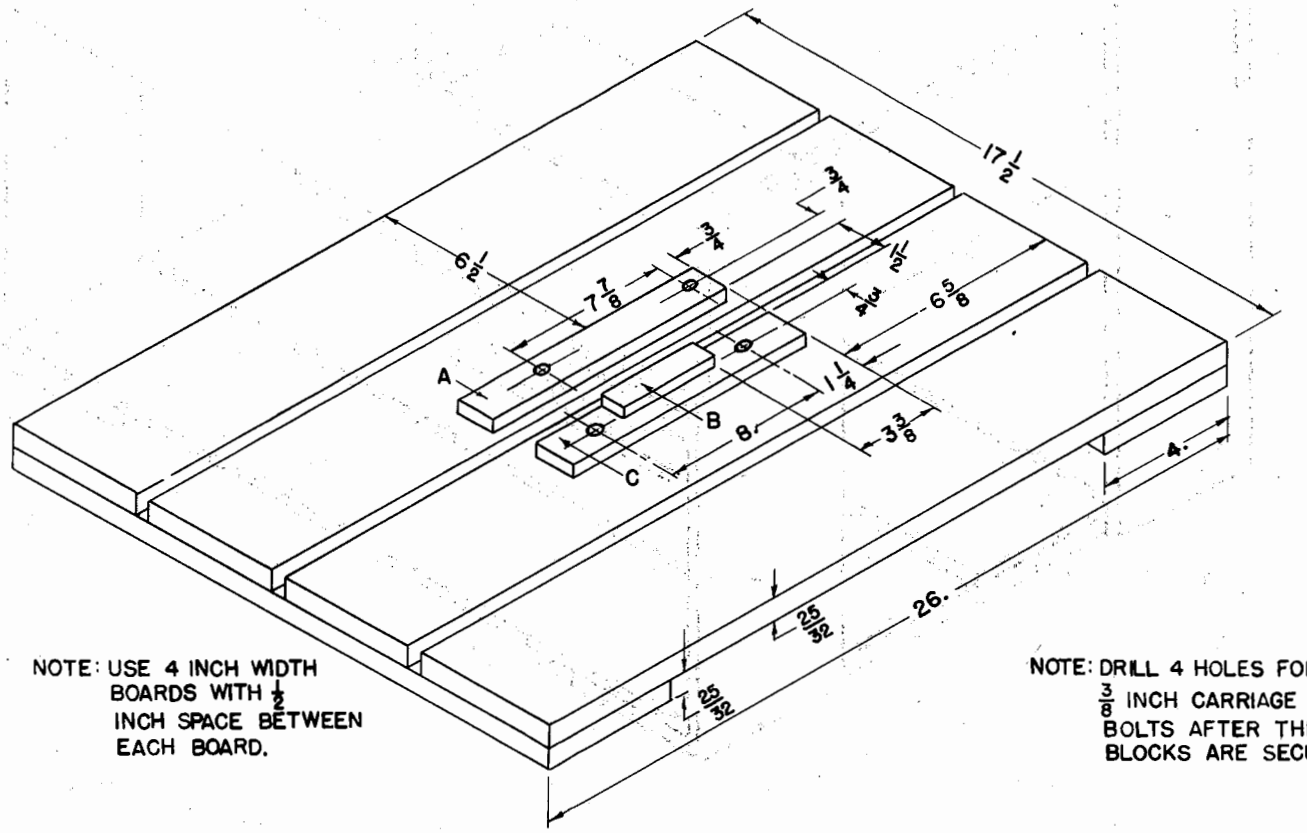


BLOCK B



BLOCK C

NOTE: HOLES ARE DRILLED WHEN BLOCKS ARE POSITIONED AND SECURED TO THE BASE.



NOTE: USE 4 INCH WIDTH BOARDS WITH $\frac{1}{2}$ INCH SPACE BETWEEN EACH BOARD.

NOTE: DRILL 4 HOLES FOR $\frac{3}{8}$ INCH CARRIAGE BOLTS AFTER THE BLOCKS ARE SECURED.

BASE FOR BALLISTIC DRIVE T23E1

FIGURE 1

Neg.#23064-1 10-10-52 FRANKFORD ARSENAL ORD. CORPS

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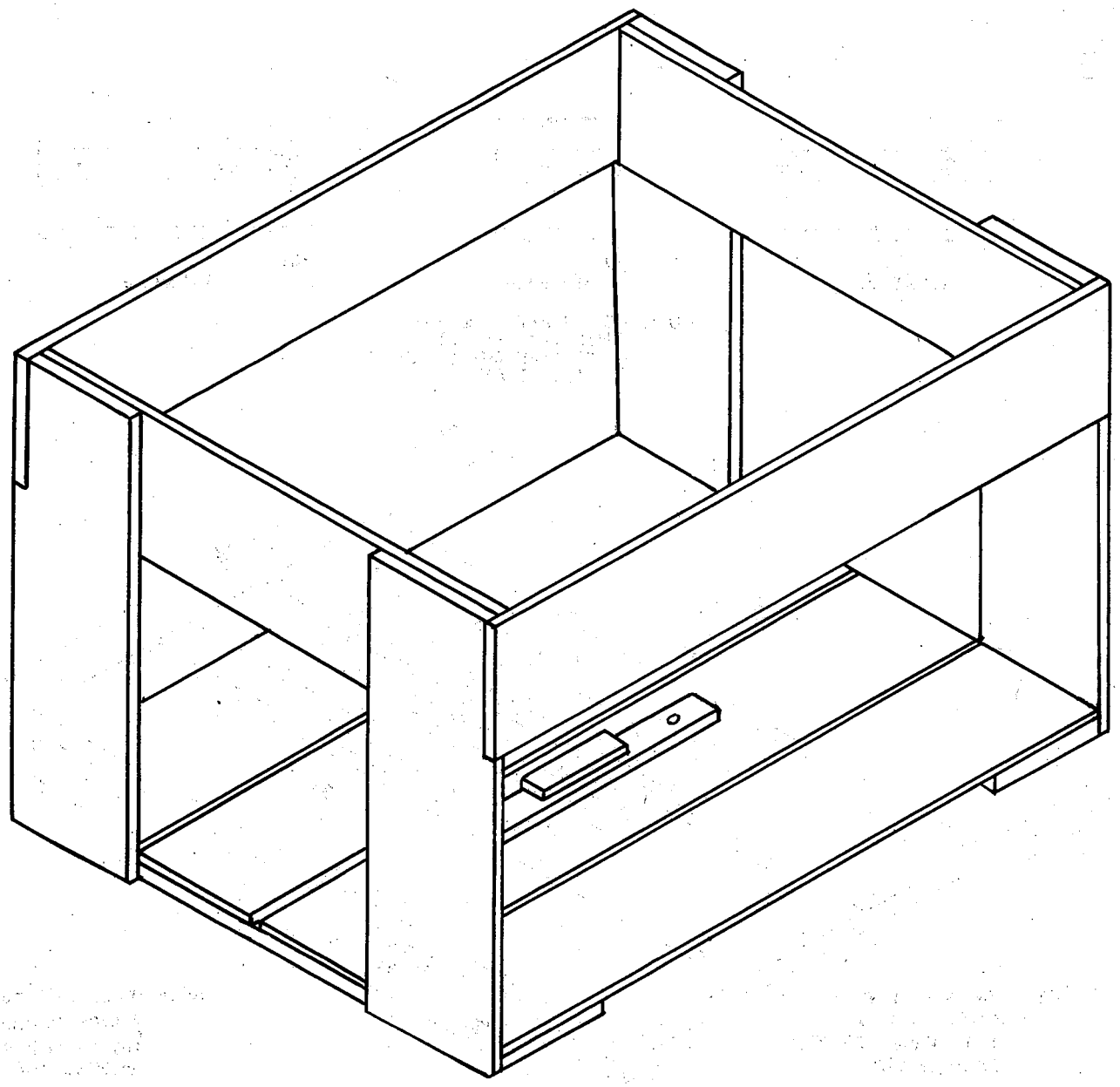
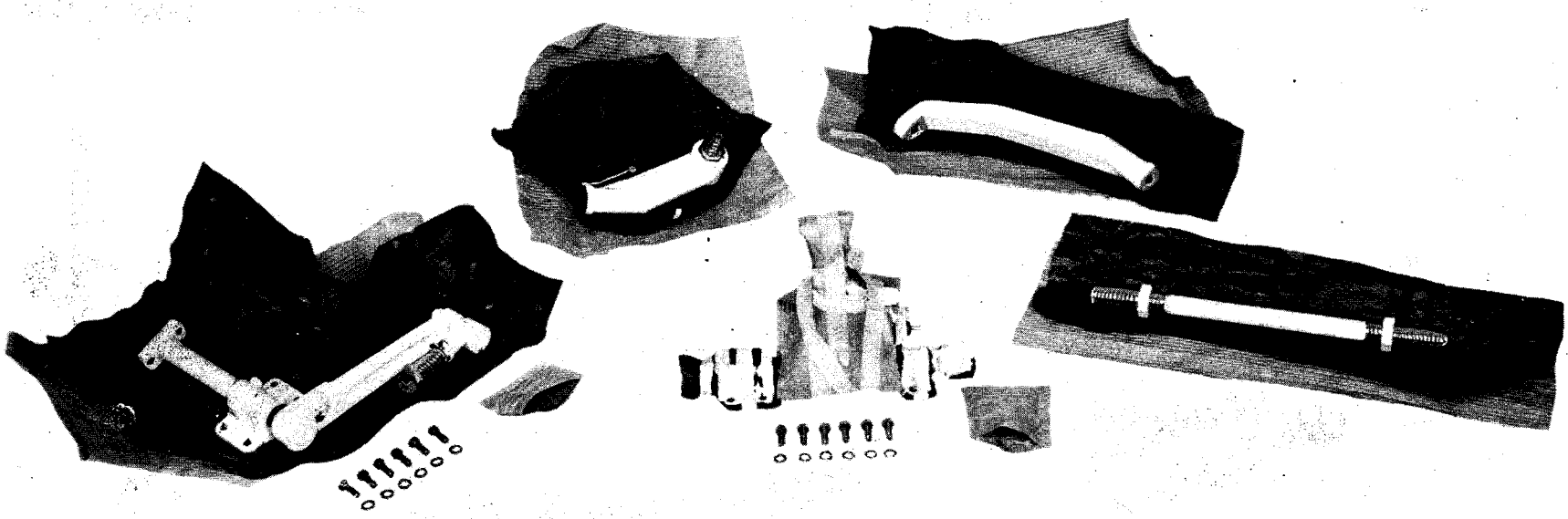


FIGURE 2

Neg.#23064-2 10-10-52 FRANKFORD ARSENAL ORD. CORPS

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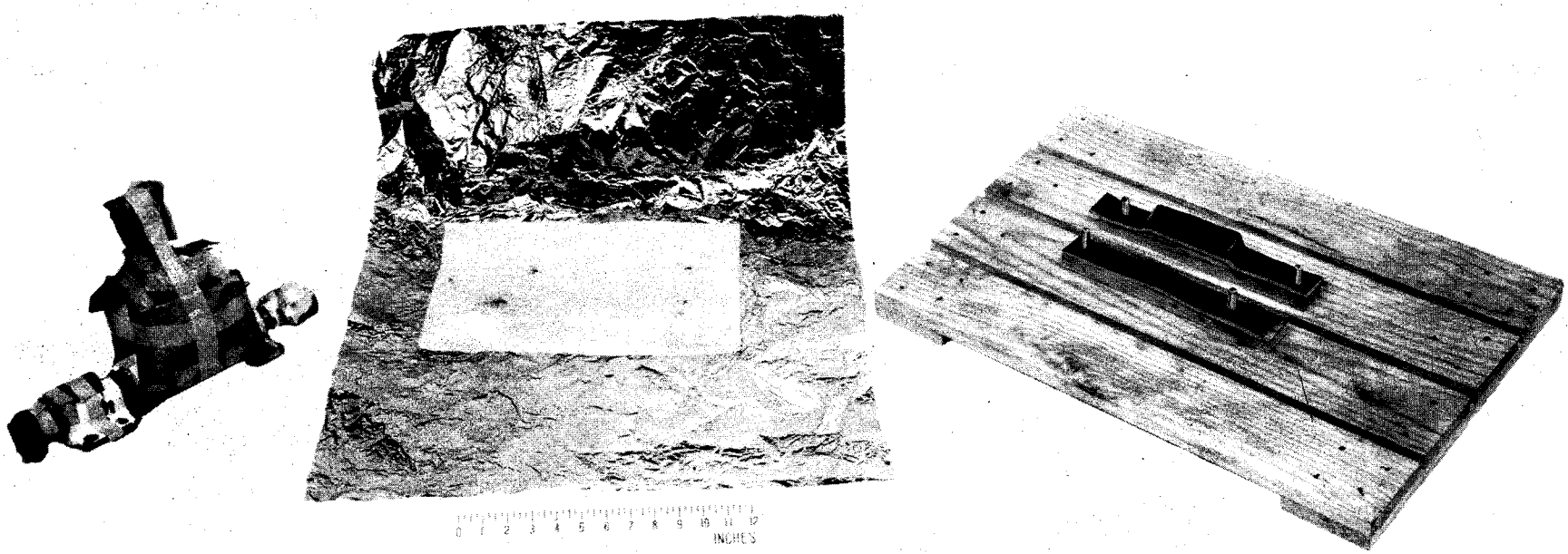


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DRIVE, BALLISTIC, T23E1 - Prepackaging the ballistic drive.

FIGURE 3

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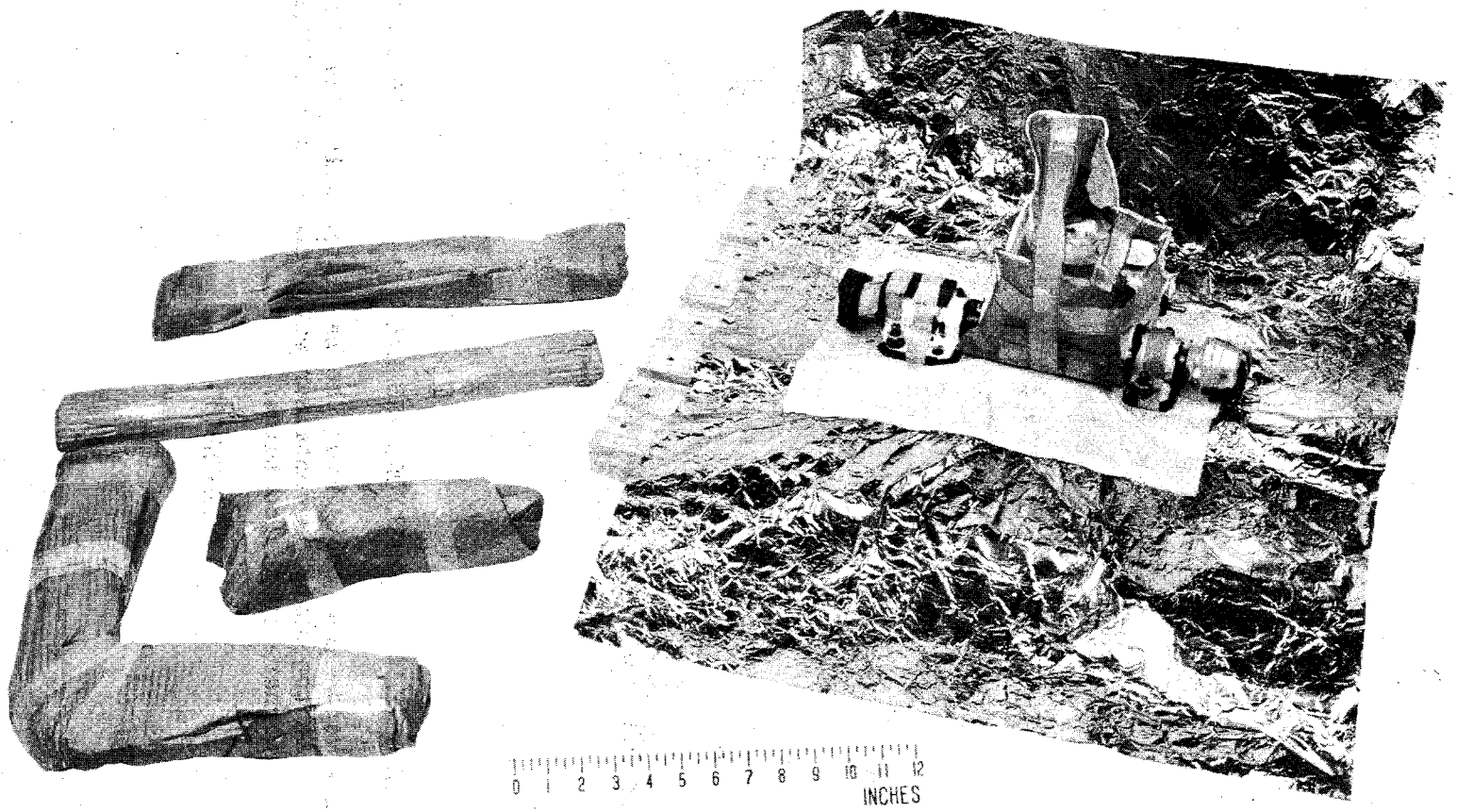


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DRIVE, BALLISTIC, T23E1 - Prepared ballistic drive, water vapor-proof bag material, and base with blocks and gaskets in place.

FIGURE 4

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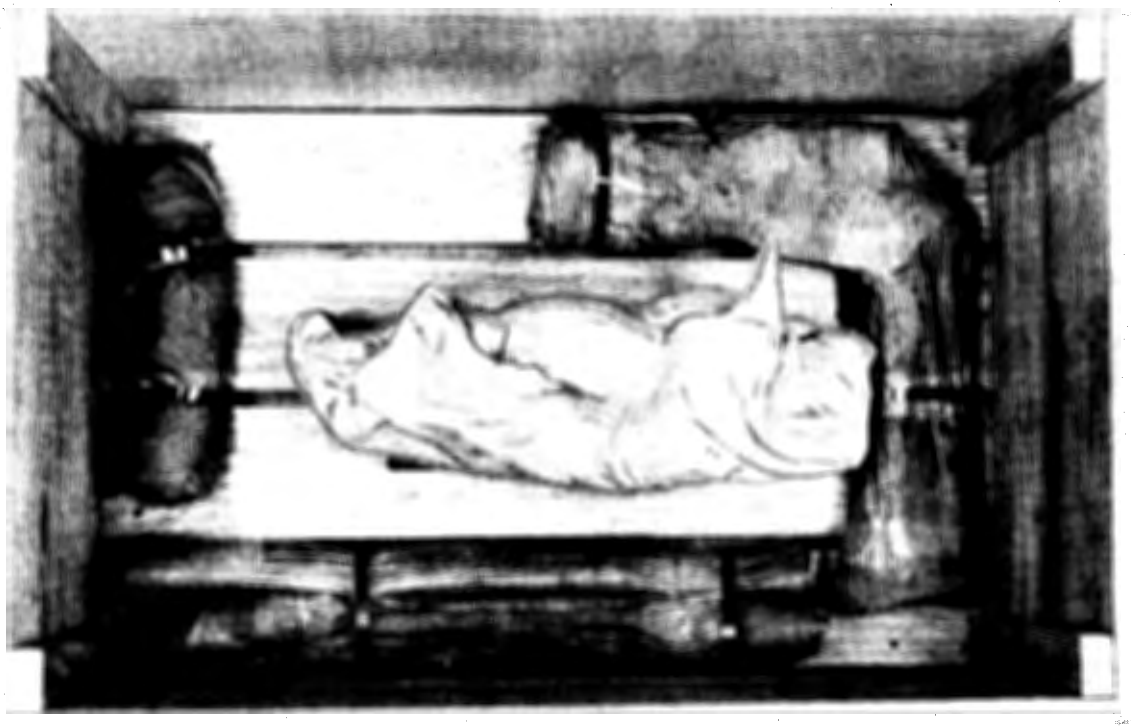


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DRIVE, BALLISTIC, T23E1 - Packaged Arms, Connector, etc., and ballistic drive mounted on base prior to sealing bag.

FIGURE 5

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DRIVE, BALLISTIC, T23E1 - Wood base, containing the ballistic drive, immobilized in the shipping container with wood cleats in the corners of the container.

FIGURE 6

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